# Time-Frequency Phase-Shifted Pilots for Massive MIMO-OFDM Channel Estimation

Jinke Tang*†, Xiqi Gao*†, Li You*†, Ding Shi*†, Jiyuan Yang*†, Xiang-Gen Xia‡,
Xinwei Zhao§, and Peigang Jiang§
*National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China
†Purple Mountain Laboratories, Nanjing 211100, China
‡Departement of Electrical and Computer Engineering, University of Delaware, Newark, DE 19716, USA
§Huawei Technologies Co., Ltd., Shanghai 210206, China
Email: {jktang, xqgao, lyou, shiding, jyyang}@seu.edu.cn, xxia@ee.udel.edu, {zhaoxinwei, jiangpeigang}@huawei.com

GLOBECOM 2024 - 2024 IEEE Global Communications Conference | 979-8-3503-5125-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/GLOBECOM52923.2024.10901171

***Abstract*—In this paper, we propose time-frequency phase-shifted pilots (TFPSPs) for massive multi-input multi-output orthogonal frequency division multiplexing (MIMO-OFDM) channel estimation. We first present a triple-beam (TB) based channel model, establishing the relationship between the space-frequency-time (SFT) domain channel and the TB domain channel. By leveraging the specific characteristics of TB domain channels, we develop TFPSPs, where distinct pilot signals are simultaneously transmitted in the frequency and time domains. Then, we present the optimal condition on TFPSP, indicating that the optimal channel estimation performance can be achieved if the TB domain channel power distributions of different UTs do not overlap with each other by scheduling TFPSPs properly. Based on this optimal condition, we propose a low-complexity pilot scheduling algorithm. Simulation results demonstrate that, compared with conventional pilot design approaches, the proposed TFPSP approach effectively improves the accuracy of channel estimation, particularly in scenarios involving a significant number of UTs.**

## I. Introduction

Massive multiple-input multiple-output (MIMO), combined with orthogonal frequency division multiplexing (OFDM), has become a core technique in 5G wireless communications, and will maintain its significance in future 6G communications [1]–[3]. The performance of the massive MIMO-OFDM system is closely related to the quality of channel state information (CSI) acquisition, commonly achieved through pilot-assisted techniques. Therefore, pilot design and channel estimation have received significant attentions. A common technique for channel acquisition is to transmit phase-shifted pilots in the frequency domain and then perform channel estimation using the specific channel characteristics, especially those in the delay domain. In recent years, this technique has undergone further refinement through its integration with the spatial-frequency (SF) beam-based channel model [4]–[6].

Regarding pilot design, phase-shifted orthogonal pilots (PSOPs) have become widely adopted for channel acquisition [6]. However, the specific format of PSOPs results in pilot overhead proportional to the number of UTs, posing a bottleneck for the system [7]. To address this issue, adjustable phase-shifted pilots (APSPs) have been proposed [8], enabling the utilization of more available pilots without increasing pilot overhead. Meanwhile, by exploiting the channel sparsity in the SF beam domain, APSPs can achieve the equivalent performance to PSOPs under specific conditions [8]. For future 6G communications, the system might support 10-100 times higher connection density and UT capacity compared to 5G [2], and conventional pilot design might also introduce considerable pilot interference. Thus, new and better pilot designs are needed.

To do so, we propose time-frequency phase-shifted pilots (TFPSPs) for massive MIMO-OFDM channel estimation. The rest of the paper is organized as follows. In Section II, we present the TB-based channel model. In Section III, we propose the TFPSP design. In Section IV, we present the optimal condition on TFPSPs and propose a pilot scheduling algorithm. Simulation results are shown in Section V, and the paper is concluded in Section VI.

## II. System Model

### A. System Configuration

We consider a single-cell TDD massive MIMO-OFDM system operating at the carrier frequency $f_{\rm c}$, where the base station (BS) is equipped with an $M$ antenna uniform linear array (ULA) and serves $U$ single-antenna users terminals (UTs). With $\lambda_{\rm c}$ as the wavelength, the inter-antenna spacing $d$ of the ULA at the BS is $0.5\lambda_{\rm c}$. Let $\mathcal{U} = \{0, 1, \cdots, U-1\}$ denote the UT set. As for the OFDM modulation, $N_{\rm c}$ and $N_{\rm g}$ denote the number of subcarriers and the length of the cyclic prefix (CP), respectively. With the sampling interval length denoted by $T_{\rm s}$, the subcarrier spacing is $\Delta f = \frac{1}{N_{\rm c}T_{\rm s}}$, and the duration of each OFDM symbol is $T_{\rm sym} = (N_{\rm c} + N_{\rm g})T_{\rm s}$. Within each OFDM symbol, we use $\mathcal{K} = \{k_0, k_0+1, \cdots, k_0+K-1\}$

This work was supported by the Jiangsu Province Basic Research Project under Grant BK20192002, the Jiangsu Province Major Science and Technology Project under Grant SBG2024000080, the Fundamental Research Funds for the Central Universities under Grants 2242022k60007 and 2242023K5003, the Key R&D Plan of Jiangsu Province under Grants BE2022067 and BE2022067-5, the National Natural Science Foundation of China under Grant 62322104, the Natural Science Foundation of Jiangsu Province under Grant BK20231415, the Natural Science Foundation on Frontier Leading Technology Basic Research Project of Jiangsu under Grant BK20222001, and the Huawei Cooperation Project.



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.

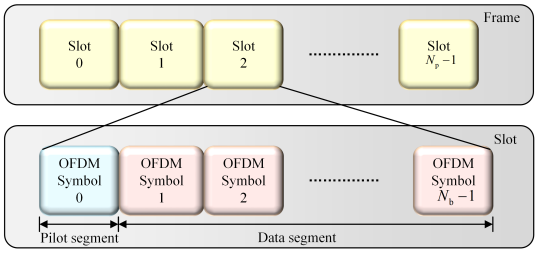



Fig. 1. Frame structure for massive MIMO-OFDM transmissions

as the index set of the $K$ valid subcarriers used for training and daY3NBM7RDCY8ta transmissions, and $\mathcal{K}$ is a subset of $\{0, 1, \cdots, N_{\rm c}-1\}$.

The time resource is divided into frames, and each frame is divided into $N_{\rm p}$ slots, as shown in Fig. 1. Each slot contains $N_{\rm b}$ OFDM symbols, where the first symbol in each slot is used for uplink (UL) pilot transmission and the other symbols are used for UL or downlink (DL) data transmissions. We can combine the current slot with its previous $(N_{\rm p}-1)$ slots to form a frame for channel estimation [9]. We assume that at the time slot $T$, the current frame for channel estimation is formed from the $(T-N_{\rm p}+1)$-th to the $T$-th slot. In this frame, with $n_T \triangleq (T-N_{\rm p}+1)$, we can define $\mathcal{P}_T \triangleq \{n_T N_{\rm b},\ n_T N_{\rm b}+1,\ \cdots,\ (T+1)\,N_{\rm b}\ -1\}$ as the OFDM symbol set, and a subset of $\mathcal{P}_T$ as the OFDM symbol set of pilot segments, i.e., $\mathcal{P}_T^{\rm p} \triangleq \{n_T N_{\rm b}, (n_T+1)\,N_{\rm b}, \cdots, TN_{\rm b}\}$. There are $N_{\rm s}=N_{\rm b}N_{\rm p}$ OFDM symbols in this frame totally, and $N_{\rm p}$ pilot segments, each of which corresponds to one OFDM symbol.

### *B. TB-based Channel Model*

Based on the above system configuration, we propose the corresponding channel model. Consider a propagation environment where there are $P_u$ paths between the BS and the $u$-th UT. The equivalent channel impulse response between the $m$-th antenna at the BS and the $u$-th UT can be expressed as [10]

$$\widetilde{h}_{u,m}\left(t,\tau\right)=\sum_{p=0}^{P_u-1}\alpha_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}t}e^{-\bar{\jmath}2\pi m\vartheta_{u,p}}\delta\left(\tau-\tau_{u,p}\right), \quad (1)$$

where $u\in\mathcal{U}$, $m\in\{0,1,\cdots,M-1\}$. $\alpha_{u,p}$, $\nu_{u,p}$, $\vartheta_{u,p}$, $\tau_{u,p}$ represent the complex-valued channel gain, the Doppler frequency, the directional cosine and the delay of the $p$-th path, respectively. $\vartheta_{u,p}$ satisfies the relationship $\vartheta_{u,p}\triangleq\frac{d}{\lambda_c}\cos\theta_{u,p}=0.5\cos\theta_{u,p}$ with the incidence angle $\theta_{u,p}$.

Let $x_{u,k,n}$ denote the transmitted signal of the $u$-th UT at the $k$-th subcarrier of the $n$-th OFDM symbol, where $k\in\mathcal{K}$. We assume that the channel state remains constant within an OFDM symbol and varies symbol by symbol due to the Doppler effect. Based on (1), the received signal on the $n$-th OFDM symbol at the $k$-th subcarrier can be written as

$$y_{m,k,n}=\sum_{u=0}^{U-1}h_{u,m,k,n}^{\rm SFT}x_{u,k,n}+\widetilde{z}, \quad (2)$$

where $\widetilde{z}$ is the noise, and $h_{u,m,k,n}^{\rm SFT}$ is the channel response in the spatial-frequency-time (SFT) domain between the $u$-th UT and the $m$-th antenna at the BS on the $k$-th subcarrier of the $n$-th OFDM symbol, which is given by [11]

$$\begin{aligned}h_{u,m,k,n}^{\rm SFT}&=\int\widetilde{h}_{u,m}\left(nT_{\rm sym},\tau\right)e^{-\bar{\jmath}2\pi k\Delta f\tau}d\tau\\&=\sum_{p=0}^{P_u-1}\alpha_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\rm sym}}e^{-\bar{\jmath}2\pi m\vartheta_{u,p}}e^{-\bar{\jmath}2\pi k\Delta f\tau_{u,p}}.\end{aligned} \quad (3)$$

We use $v_{\rm speed}$ to denote the maximum UT speed in the system and $\nu_{\max}\triangleq 2v_{\rm speed}/\lambda_c$. Assuming all the UTs are synchronized [5], we have $\vartheta_{u,p}\in[-0.5,0.5]$, $\tau_{u,p}\in[0,1/\Delta f]$ and $\nu_{u,p}\in[-\nu_{\max}/2,\nu_{\max}/2]$. By setting $N_{\rm b}<1/\left(T_{\rm sym}\nu_{\max}\right)$, we have that $\nu_{\max}<1/\left(N_{\rm b}T_{\rm sym}\right)$, and $\nu_{u,p}\in[-1/(2N_{\rm b}T_{\rm sym}),1/(2N_{\rm b}T_{\rm sym})]$. Define $\vartheta_n\triangleq\frac{n-N_\vartheta/2}{N_\vartheta}$, $\tau_n\triangleq\frac{n}{N_\tau\Delta f}$, $\nu_n\triangleq\frac{(n-N_\nu/2)}{N_\nu N_{\rm b}T_{\rm sym}}$, and denote the sets of the discretized grids $\vartheta_n$, $\tau_n$ and $\nu_n$ as $\mathcal{S}_\vartheta\triangleq\{\vartheta_0,\vartheta_1,\cdots,\vartheta_{N_\vartheta-1}\}$, $\mathcal{S}_\tau\triangleq\{\tau_0,\tau_1,\cdots,\tau_{N_\tau-1}\}$ and $\mathcal{S}_\nu\triangleq\{\nu_0,\nu_1,\cdots,\nu_{N_\nu-1}\}$, where $N_\vartheta$, $N_\tau$, and $N_\nu$ are the numbers of points of the grids $\vartheta_n$, $\tau_n$, and $\nu_n$ in their ranges given above, respectively. Since all the discretized parameters $\vartheta_n$, $\tau_n$ and $\nu_n$ are acquired by sampling $[-0.5,0.5]$, $[0,1/\Delta f]$ and $[-1/(2N_{\rm b}T_{\rm sym}),1/(2N_{\rm b}T_{\rm sym})]$, respectively, we call $\mathcal{S}_{\rm TB}\triangleq\mathcal{S}_\vartheta\times\mathcal{S}_\tau\times\mathcal{S}_\nu$ as the range of the triple-beam (TB) domain. When $N_\vartheta$, $N_\tau$ and $N_\nu$ are sufficiently large, we use

$$\bar{\vartheta}_{u,p}=\min_{\vartheta\in S_\vartheta}\left|\vartheta_{u,p}-\vartheta\right|, \quad (4a)$$

$$\bar{\tau}_{u,p}=\min_{\tau\in S_\tau}\left|\tau_{u,p}-\tau\right|, \quad (4b)$$

$$\bar{\nu}_{u,p}=\min_{\nu\in S_\nu}\left|\nu_{u,p}-\nu\right|, \quad (4c)$$

to approximate $\vartheta_{u,p}$, $\tau_{u,p}$ and $\nu_{u,p}$, respectively. With $\mathcal{S}_u\triangleq\{(\bar{\vartheta}_{u,0},\bar{\tau}_{u,0},\bar{\nu}_{u,0}),(\bar{\vartheta}_{u,1},\bar{\tau}_{u,1},\bar{\nu}_{u,1}),\cdots,(\bar{\vartheta}_{u,P_u-1},\bar{\tau}_{u,P_u-1},\bar{\nu}_{u,P_u-1})\}$ as the parameter set of different paths, we have $\mathcal{S}_u\subseteq\mathcal{S}_{\rm TB}$.

According to the frame structure, at the time slot $T$, the indices of all the OFDM symbols in the current frame are included in $\mathcal{P}_T$. At the BS, we define the following steering vectors

$$\mathbf{v}^{\rm s}\left(\vartheta\right)\triangleq\left[1,e^{-\bar{\jmath}2\pi\vartheta},\cdots,e^{-\bar{\jmath}2\pi(M-1)\vartheta}\right]^{\rm T}, \quad (5)$$

$$\mathbf{v}^{\rm f}\left(\tau\right)\triangleq\left[e^{-\bar{\jmath}2\pi k_0\Delta f\tau},e^{-\bar{\jmath}2\pi(k_0+1)\Delta f\tau},\cdots,e^{-\bar{\jmath}2\pi(k_0+K-1)\Delta f\tau}\right]^{\rm T}, \quad (6)$$

$$\mathbf{v}_T^{\rm t}\left(\nu\right)\triangleq e^{\bar{\jmath}2\pi n_T N_{\rm b}\nu T_{\rm sym}}.\left[1,e^{\bar{\jmath}2\pi\nu T_{\rm sym}},\cdots,e^{\bar{\jmath}2\pi\nu(N_{\rm s}-1)T_{\rm sym}}\right]^{\rm T}, \quad (7)$$

which correspond to physical beams in the spatial, frequency and time domains, directed towards the directional cosine $\vartheta$, the delay $\tau$ and the Doppler frequency $\nu$, respectively. With the proposed frame structure in Fig. 1, we can select the pilot segments from the current frame, and the time domain steering vectors of the pilot segments can then be expressed as $\mathbf{v}_T^{\rm t,p}\left(\nu\right)\triangleq\left[\mathbf{v}_T^{\rm t}\left(\nu\right)\right]_{0:N_{\rm b}:N_{\rm s}-1,:}\in\mathbb{C}^{N_{\rm p}\times 1}$, where the added superscript p stands for "pilot".

With the steering vectors, the SFT domain channel between



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.

the $u$-th UT and the BS in the pilot segments of the current frame can be expressed as an $MKN_{\mathrm{p}} \times 1$ vector, i.e,

$$\begin{aligned}\mathbf{h}_{u,T}^{\mathrm{SFT,p}} &= \sum_{p=0}^{P_u-1} \alpha_{u,p}\mathbf{v}_T^{\mathrm{t,p}}\left(\nu_{u,p}\right) \otimes \left(\mathbf{v}^{\mathrm{f}}\left(\tau_{u,p}\right) \otimes \mathbf{v}^{\mathrm{s}}(\vartheta_{u,p})\right) \\ &\approx \sum_{p=0}^{P_u-1} \alpha_{u,p}\mathbf{v}_T^{\mathrm{t,p}}\left(\bar{\nu}_{u,p}\right) \otimes \left(\mathbf{v}^{\mathrm{f}}\left(\bar{\tau}_{u,p}\right) \otimes \mathbf{v}^{\mathrm{s}}(\bar{\vartheta}_{u,p})\right), \quad (8)\end{aligned}$$

where $\otimes$ denotes the Kronecker product operation and the $(m+kM+nMK)$-th element of $\mathbf{h}_{u,T}^{\mathrm{SFT,p}}$ is $h_{u,m,k,(n_T+n)N_{\mathrm{b}}}^{\mathrm{SFT}}$.

Furthermore, with the definition

$$\alpha_{u,n_\vartheta,n_\tau,n_\nu} = \begin{cases} \sum\limits_{p:\left(\bar{\vartheta}_{u,p},\bar{\tau}_{u,p},\bar{\nu}_{u,p}\right)=\left(\vartheta_{n_\vartheta},\tau_{n_\tau},\nu_{n_\nu}\right)} \alpha_{u,p}, & \left(\vartheta_{n_\vartheta},\tau_{n_\tau},\nu_{n_\nu}\right) \in \mathcal{S}_u, \\ 0, & \left(\vartheta_{n_\vartheta},\tau_{n_\tau},\nu_{n_\nu}\right) \notin \mathcal{S}_u, \end{cases} \quad (9)$$

where $\bar{\vartheta}_{u,p}$, $\bar{\tau}_{u,p}$, $\bar{\nu}_{u,p}$ are defined in (4a)-(4c), $\mathbf{h}_{u,T}^{\mathrm{SFT,p}}$ in (8) can be rewritten as

$$\begin{aligned}\mathbf{h}_{u,T}^{\mathrm{SFT,p}} = \sum_{n_\vartheta=0}^{N_\vartheta-1}\sum_{n_\tau=0}^{N_\tau-1}\sum_{n_\nu=0}^{N_\nu-1} \alpha_{u,n_\vartheta,n_\tau,n_\nu}\mathbf{v}_T^{\mathrm{t,p}}\left(\nu_{n_\nu}\right) \\ \otimes \left(\mathbf{v}^{\mathrm{f}}\left(\tau_{n_\tau}\right) \otimes \mathbf{v}^{\mathrm{s}}(\vartheta_{n_\vartheta})\right). \quad (10)\end{aligned}$$

By packing all the sampled steering vectors $\mathbf{v}^{\mathrm{s}}(\vartheta_{n_\vartheta})$, $\mathbf{v}^{\mathrm{f}}\left(\tau_{n_\tau}\right)$ and $\mathbf{v}_T^{\mathrm{t,p}}\left(\nu_{n_\nu}\right)$ into beam matrices $\mathbf{V}^{\mathrm{s}} \triangleq [\mathbf{v}^{\mathrm{s}}(\vartheta_0), \mathbf{v}^{\mathrm{s}}(\vartheta_1), \cdots, \mathbf{v}^{\mathrm{s}}(\vartheta_{N_\vartheta-1})] \in \mathbb{C}^{M\times N_\vartheta}$, $\mathbf{V}^{\mathrm{f}} \triangleq [\mathbf{v}^{\mathrm{f}}(\tau_0), \mathbf{v}^{\mathrm{f}}(\tau_1), \cdots, \mathbf{v}^{\mathrm{f}}(\tau_{N_\tau-1})] \in \mathbb{C}^{K\times N_\tau}$ and $\mathbf{V}_T^{\mathrm{t,p}} \triangleq [\mathbf{v}_T^{\mathrm{t,p}}(\nu_0), \mathbf{v}_T^{\mathrm{t,p}}(\nu_1), \cdots, \mathbf{v}_T^{\mathrm{t,p}}(\nu_{N_\nu-1})] \in \mathbb{C}^{N_{\mathrm{p}}\times N_\nu}$, the expression in (10) becomes

$$\mathbf{h}_{u,T}^{\mathrm{SFT,p}} = \underbrace{\mathbf{V}_T^{\mathrm{t,p}} \otimes \left(\mathbf{V}^{\mathrm{f}} \otimes \mathbf{V}^{\mathrm{s}}\right)}_{\triangleq \mathbf{V}_T^{\mathrm{p}}} \mathbf{h}_u^{\mathrm{TB}}, \quad (11)$$

where $\mathbf{h}_u^{\mathrm{TB}} \in \mathbb{C}^{N_\vartheta N_\tau N_\nu \times 1}$ whose $(n_\vartheta + n_\tau N_\vartheta + n_\nu N_\vartheta N_\tau)$-th element is $\alpha_{u,n_\vartheta,n_\tau,n_\nu}$. We call $\mathbf{h}_u^{\mathrm{TB}}$ as the TB domain channel vector.

In the following, we use $\mathbf{F}_A$ to denote the $A \times A$ discrete Fourier transform (DFT) matrix and $\mathbf{f}_{A,a}$ as its $\langle a \rangle_A$-th column, where $\langle\cdot\rangle_A$ denotes the modulo-$A$ operation. Besides, we define the following matrix

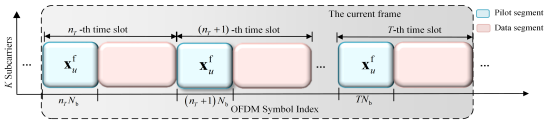
$$\mathbf{\Gamma}_{A,a} \triangleq \begin{bmatrix} \mathbf{0} & \mathbf{I}_{A-\langle a\rangle_A} \\ \mathbf{I}_{\langle a\rangle_A} & \mathbf{0} \end{bmatrix}. \quad (12)$$

Then, with $\mathbf{F}_{A,g} \triangleq \mathbf{F}_A\mathbf{\Gamma}_{A,g}$, when we set $N_\vartheta = F_\vartheta M$, $N_\tau = F_\tau K$, $N_\nu = F_\nu N_{\mathrm{p}}$, and the fine factors $F_\vartheta, F_\tau, F_\nu \geq 1$, $\mathbf{V}^{\mathrm{s}}$, $\mathbf{V}^{\mathrm{f}}$ and $\mathbf{V}_T^{\mathrm{t,p}}$ can all be generated from the DFT matrices or partial DFT matrices, i.e., $\mathbf{V}^{\mathrm{s}} = \left[\mathbf{F}_{N_\vartheta,N_\vartheta/2}\right]_{0:M-1,:}$, $\mathbf{V}^{\mathrm{f}} = [\mathbf{F}_{N_\tau}]_{\mathcal{K},:}$ and $\mathbf{V}_T^{\mathrm{t,p}} = [\mathbf{\Gamma}_{N_\nu,n_T}\mathbf{F}^*_{N_\nu,N_\nu/2}]_{0:N_{\mathrm{p}}-1,:}$.

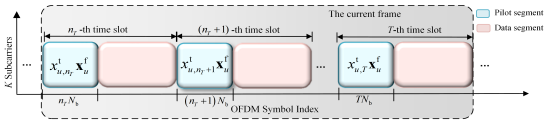
From (8)-(9), the TB domain channel is supported on set $\mathcal{S}_u$, i.e., its components are only non-zero on $\mathcal{S}_u$. Based on the OFDM design principle, the channel delay spread is relatively small to $1/\Delta f$. Similarly, in the Doppler domain, the Doppler spread width $\nu_{\max}$ is relatively small to $1/(N_{\mathrm{b}}T_{\mathrm{sym}})$. Therefore, $\mathcal{S}_u$ should be a small subset of $\mathcal{S}_{\mathrm{TB}}$, and $\mathbf{h}_u^{\mathrm{TB}}$ is a sparse vector. Then we define the TB domain channel covariance matrix $\mathbf{R}_u^{\mathrm{TB}} \triangleq \mathbb{E}\left\{\mathbf{h}_u^{\mathrm{TB}}\left(\mathbf{h}_u^{\mathrm{TB}}\right)^{\mathrm{H}}\right\}$, which is a diagonal matrix and the $(n_\vartheta + n_\tau N_\vartheta + n_\nu N_\vartheta N_\tau)$-th diagonal element is the power of the $(n_\vartheta, n_\tau, n_\nu)$-th beam in the TB domain. Accordingly, in the SFT domain, we can also define the SFT domain channel covariance matrix $\mathbf{R}_{u,T}^{\mathrm{SFT,p}} \triangleq \mathbb{E}\left\{\mathbf{h}_{u,T}^{\mathrm{SFT,p}}\left(\mathbf{h}_{u,T}^{\mathrm{SFT,p}}\right)^{\mathrm{H}}\right\}$. With the relationship in (11), $\mathbf{R}_{u,T}^{\mathrm{SFT,p}}$ can be calculated from $\mathbf{R}_u^{\mathrm{TB}}$ by

$$\mathbf{R}_{u,T}^{\mathrm{SFT,p}} = \mathbf{V}_T\mathbf{R}_u^{\mathrm{TB}}\mathbf{V}_T^{\mathrm{H}}. \quad (13)$$

With the channel sparsity in the TB domain, the number of non-zero elements in $\mathbf{R}_u^{\mathrm{TB}}$ is much smaller than $\mathbf{R}_{u,T}^{\mathrm{SFT,p}}$. Meanwhile, as a kind of statistical CSI, $\mathbf{R}_u^{\mathrm{TB}}$ varies slowly with time. Hence, abundant resources can be adopted for acquiring $\mathbf{R}_u^{\mathrm{TB}}$ [9], and we assume that $\mathbf{R}_u^{\mathrm{TB}}$ of all the UTs are available at the BS in the rest of this paper.




(a) FPSPs transmitted in the current frame at the time slot $T$.




(b) TFPSPs transmitted in the current frame at the time slot $T$.

Fig. 2. Pilots transmitted in the current frame at the time slot $T$.

## III. Time-frequency Phase-shifted Pilot

In this section, we propose our TFPSP design based on the TB-based channel model. The idea can be viewed as an extension of the popular phase-shifted pilots in the frequency domain to the time-frequency domain.

We first apply the widely-used frequency phase-shifted pilots (FPSPs) to the proposed channel model. With $\mathbf{x}_u^{\mathrm{f}} \in \mathbb{C}^{K\times 1}$ as the pilot sequence transmitted from the $u$-th UT and $[\mathbf{x}_u^{\mathrm{f}}]_k$ as the signal transmitted on the $(k+k_0)$-th subcarrier, $\mathbf{x}_u^{\mathrm{f}}$ can be designed as

$$\mathbf{x}_u^{\mathrm{f}} = \mathbf{f}_{\phi_u}^{\mathrm{f}} \odot \mathbf{x}^{\mathrm{f}}, \quad (14)$$

where $\odot$ denotes the point-wise product operation, $\phi_u \in \{0, 1, \cdots, K-1\}$ represents the phase shift of the $u$-th UT, and $\mathbf{f}_{\phi_u}^{\mathrm{f}} = \left[e^{-\bar{\jmath}2\pi k_0\frac{\phi_u}{K}}, e^{-\bar{\jmath}2\pi(k_0+1)\frac{\phi_u}{K}}, \cdots, e^{-\bar{\jmath}2\pi(k_0+K-1)\frac{\phi_u}{K}}\right]$, which can be generated from the $K\times K$ DFT matrix, i.e., $\mathbf{f}_{\phi_u}^{\mathrm{f}} = e^{-\bar{\jmath}2\pi k_0\frac{\phi_u}{K}}\mathbf{f}_{K,\phi_u}$. $\mathbf{x}^{\mathrm{f}}$ is a basic pilot sequence shared by all UTs, whose components are all unit modulus, and a Zadoff-Chu (ZC) sequence is usually used to generate $\mathbf{x}^{\mathrm{f}}$ [12]. Both PSOPs and APSPs belong to FPSPs, and they differ in the value of $\phi_u$. Applying FPSPs to the frame structure in Fig. 1, at the time slot $T$, all the OFDM symbols in the pilot segments of the current frame transmit the same $\mathbf{x}_u^{\mathrm{f}}$ over the valid subcarriers, as shown in Fig. 2 (a).



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.

The wide application of FPSP is related to the channel characteristics in the delay domain [8]. Meanwhile, similar channel characteristics can also be exploited in the Doppler domain, where typically channel sparsity exists. Accordingly, phase shift pilots can also be transmitted in the time domain. In this case, as illustrated in Fig. 2 (b), $\mathbf{x}_u^{\mathrm{f}}$ transmitted in the pilot segment of the $n$-th slot is multiplied by $x_{u,n}^{\mathrm{t}}$, which may be different for different $n$. Then along the time dimension, $x_{u,0}^{\mathrm{t}}, x_{u,1}^{\mathrm{t}}, x_{u,2}^{\mathrm{t}}, \cdots$ constitute the time domain pilot sequence. Similar to the design of $\mathbf{x}_u^{\mathrm{f}}$ in (14), $x_{u,n}^{\mathrm{t}}$ can also be generated by a phase shift sequence and a ZC sequence, i.e.,

$$x_{u,n}^{\mathrm{t}} = e^{-\bar{\jmath}2\pi n\frac{\varphi_u}{N_{\mathrm{p}}}} \cdot e^{\bar{\jmath}2\pi\gamma\frac{n\left(n+\langle N_{\mathrm{p}}\rangle_2\right)}{N_{\mathrm{p}}}}, n = 0, 1, 2, \cdots, \quad (15)$$

where $\gamma$ is an integer that is coprime to $N_{\mathrm{p}}$ [12], and the phase shift $\varphi_u \in \{0, 1, \cdots, N_{\mathrm{p}} - 1\}$. Under this setting, we have $x_{u,n}^{\mathrm{t}} = x_{u,n+N_{\mathrm{p}}}^{\mathrm{t}}$, indicating that $x_{u,n}^{\mathrm{t}}$ is a periodic sequence with period $N_{\mathrm{p}}$. At the time slot $T$, the pilot signal of the current frame is $\mathbf{x}_{u,T}^{\mathrm{t}} \triangleq [x_{u,n_T}^{\mathrm{t}}, x_{u,n_T+1}^{\mathrm{t}}, \cdots, x_{u,T}^{\mathrm{t}}]^{\mathrm{T}}$. Due to the periodicity of $x_{u,n}^{\mathrm{t}}$, we can express $\mathbf{x}_{u,T}^{\mathrm{t}}$ as

$$\mathbf{x}_{u,T}^{\mathrm{t}} = \mathbf{\Gamma}_{N_{\mathrm{p}},n_T}\left(\mathbf{f}_{\varphi_u}^{\mathrm{t}} \odot \mathbf{x}^{\mathrm{t}}\right), \quad (16)$$

where $\mathbf{f}_{\varphi_u}^{\mathrm{t}} = \mathbf{f}_{N_{\mathrm{p}},\varphi_u}$ and $\mathbf{x}^{\mathrm{t}}$ is composed of the $N_{\mathrm{p}}$-point ZC sequence with $[\mathbf{x}^{\mathrm{t}}]_n = e^{\bar{\jmath}2\pi\gamma n\left(n+\langle N_{\mathrm{p}}\rangle_2\right)/N_{\mathrm{p}}}$. With the above settings, signals transmitted on different pilot segments may vary, and we refer to such a design as TFPSP.

At the time slot $T$, by transmitting TFPSPs, the received signal at the BS can be expressed as

$$\mathbf{y}_T^{\mathrm{p}} = \sqrt{\sigma_{\mathrm{p}}}\sum_{u=0}^{U-1}\underbrace{\left(\mathbf{X}_{u,T}^{\mathrm{t}} \otimes \left(\mathbf{X}_u^{\mathrm{f}} \otimes \mathbf{I}_M\right)\right)}_{\triangleq \mathbf{X}_{u,T}}\mathbf{h}_{u,T}^{\mathrm{SFT,p}} + \mathbf{z}, \quad (17)$$

where $\mathbf{X}_u^{\mathrm{f}} \triangleq \mathrm{diag}\{\mathbf{x}_u^{\mathrm{f}}\}$, $\mathbf{X}_{u,T}^{\mathrm{t}} = \mathrm{Diag}\left\{\mathbf{x}_{u,T}^{\mathrm{t}}\right\}$, $\sigma_{\mathrm{p}}$ is the signal transmit power and $\mathbf{z}$ is the additive white Gaussian noise (AWGN) vector with each element identically and independently distributed as $\mathcal{CN}(0, \sigma_z)$. Specially, transmitting FPSPs is equivalent to setting $\mathbf{X}_{u,T}^{\mathrm{t}} = \mathbf{I}_{N_{\mathrm{p}}}$ in (17), or $\varphi_u = 0$ and $\mathbf{x}_{u,T}^{\mathrm{t}}$ in (16) is the all 1 vector $\mathbf{1}$ for all $u \in \mathcal{U}$.

## IV. Channel Estimation with TFPSPs

For TFPSPs, both $\phi_u$ and $\varphi_u$ can be scheduled, and this may have an impact on channel estimation performance as we shall see below.

With the relationship in (11), the received signal in (17) can be further expressed with $\mathbf{h}_u^{\mathrm{TB}}$ as

$$\mathbf{y}_T^{\mathrm{p}} = \sqrt{\sigma_{\mathrm{p}}}\mathbf{X}_{u,T}\mathbf{V}_T^{\mathrm{p}}\mathbf{h}_u^{\mathrm{TB}} + \sqrt{\sigma_{\mathrm{p}}}\sum_{u'\neq u}\mathbf{X}_{u',T}\mathbf{V}_T^{\mathrm{p}}\mathbf{h}_{u'}^{\mathrm{TB}} + \mathbf{z}. \quad (18)$$

Then by adopting the criterion of MMSE, for each UT, the estimate of $\mathbf{h}_u^{\mathrm{TB}}$ is given as

$$\widehat{\mathbf{h}}_u^{\mathrm{TB}} = \mathbf{R}_u^{\mathrm{TB}}(\mathbf{X}_{u,T}\mathbf{V}_T^{\mathrm{p}})^{\mathrm{H}}\mathbf{C}_{\mathrm{all}}^{-1}\frac{\mathbf{y}_T^{\mathrm{p}}}{\sqrt{\sigma_{\mathrm{p}}}}, \quad (19)$$

where $\mathbf{C}_u \triangleq \mathbf{X}_{u,T}\mathbf{R}_{u,T}^{\mathrm{SFT,p}}\mathbf{X}_{u,T}^{\mathrm{H}}$ and $\mathbf{C}_{\mathrm{all}} \triangleq \sum_{u=0}^{U-1}\mathbf{C}_u + \sigma_z\mathbf{I}_{MKN_{\mathrm{p}}}$. With the relationship in (11), the MMSE estimate of $\mathbf{h}_{u,T}^{\mathrm{SFT,p}}$ is

$$\hat{\mathbf{h}}_{u,T}^{\mathrm{SFT,p}} = \mathbf{R}_{u,T}^{\mathrm{SFT,p}}\mathbf{X}_{u,T}^{\mathrm{H}}\mathbf{C}_{\mathrm{all}}^{-1}\frac{\mathbf{y}_T^{\mathrm{p}}}{\sqrt{\sigma_{\mathrm{p}}}} \quad (20)$$

and the MSE between $\mathbf{h}_{u,T}^{\mathrm{SFT,p}}$ and $\hat{\mathbf{h}}_{u,T}^{\mathrm{SFT,p}}$ is then given by

$$\varepsilon = \sum_{u=0}^{U-1}\mathrm{tr}\left\{\mathbf{R}_{u,T}^{\mathrm{SFT,p}} - \mathbf{R}_{u,T}^{\mathrm{SFT,p}}\mathbf{X}_{u,T}^{\mathrm{H}}\mathbf{C}_{\mathrm{all}}^{-1}\mathbf{X}_{u,T}\mathbf{R}_{u,T}^{\mathrm{SFT,p}}\right\}. \quad (21)$$

It can be observed from (21) that, the value of $\varepsilon_{\mathrm{MSE}}$ depends on the statistical CSI of each UT, as well as the interference among all the UTs in $\mathcal{U}$, which is included in $\mathbf{C}_{\mathrm{all}}$. In order to eliminate the inter-UT interference and reduce the MSE, we make the following analysis. According to the pilot design, we can establish the following relationships $\mathbf{X}_u^{\mathrm{f}}\mathbf{V}^{\mathrm{f}} = \mathbf{X}^{\mathrm{f}}\mathbf{V}^{\mathrm{f}}\mathbf{\Gamma}_{N_\tau,-\phi_u F_\tau}$ and $\mathbf{X}_{u,T}^{\mathrm{t}}\mathbf{V}_T^{\mathrm{t,p}} = \mathbf{X}_T^{\mathrm{t}}\mathbf{V}_T^{\mathrm{t,p}}\mathbf{\Gamma}_{N_\nu,\varphi_u F_\nu}$. In this case, $\mathbf{C}_u$ can be further expressed as

$$\mathbf{C}_u = \mathbf{X}_T\mathbf{V}_T^{\mathrm{p}}\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}(\mathbf{X}_T\mathbf{V}_T^{\mathrm{p}})^{\mathrm{H}} \quad (22)$$

where $L_{\phi_u} = -\phi_u F_\tau$, $L_{\varphi_u} = \varphi_u F_\nu$ and

$$\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}} = \left(\mathbf{\Gamma}_{N_\nu,L_{\varphi_u}} \otimes \left(\mathbf{\Gamma}_{N_\tau,L_{\phi_u}} \otimes \mathbf{I}_M\right)\right)\mathbf{R}_u^{\mathrm{TB}}. \quad (23)$$

Actually, $\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}$ does not change the values of diagonal elements in $\mathbf{R}_u^{\mathrm{TB}}$, and it just changes their positions. Besides, different phase factors are equivalent to different positional changes. Therefore, we call $\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}$ as the equivalent TB domain channel covariance matrix, and the diagonal of $\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}$ as the the equivalent TB domain power distribution. On this basis, we have the following theorem, giving the minimum value of MSE and the condition of reaching the value.

*Theorem 1:* When $M, K, N_{\mathrm{p}} \to \infty$, $\varepsilon_{\mathrm{MSE}}$ is lower bounded by

$$\varepsilon_{\min} = \sum_{u=0}^{U-1}\mathrm{tr}\left\{\mathbf{R}_{u,T}^{\mathrm{SFT,p}} - \mathbf{R}_{u,T}^{\mathrm{SFT,p}}\mathbf{X}_{u,T}^{\mathrm{H}}\left(\mathbf{C}_u + \sigma_z\mathbf{I}_{MKN_{\mathrm{p}}}\right)^{-1}\mathbf{X}_{u,T}\mathbf{R}_{u,T}^{\mathrm{SFT,p}}\right\}. \quad (24)$$

This lower bound can be achieved if $\forall u, u' \in \mathcal{U}$, $u \neq u'$,

$$\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}\mathbf{R}_{u',\phi_u',\varphi_u'}^{\mathrm{TB}} = \mathbf{0}. \quad (25)$$

*Proof*: The proof can be found in the long version of this paper [9].

The condition in (25) implys that the equivalent channel power distributions of different UTs are non-overlapping in the TB domain. Compared with $\varepsilon_{\mathrm{MSE}}$ in (21), the inter-UT interference is eliminated in $\varepsilon_{\mathrm{MSE,min}}$. Hence, Theorem 1 suggests that non-overlapping equivalent channel power distributions in the TB domain lead to the elimination of inter-UT interference and the minimization of $\varepsilon_{\mathrm{MSE}}$. Since $\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}$ for each $u \in \mathcal{U}$ is determined by the phase shifts $(\phi_u, \varphi_u)$ of TFPSPs, the condition (25) can be achieved by selecting proper TFPSPs for all $u \in \mathcal{U}$. Meanwhile, with the channel sparsity, only a few diagonal elements of $\mathbf{R}_{u,\phi_u,\varphi_u}^{\mathrm{TB}}$ is dominating, and even the same TFPSP can be scheduled to UTs with no overlap in the TB domain equivalent channels.

In practical systems, as the number of UTs increases, (25)



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.

may not always be well satisfied. However, we can still follow such an idea and schedule TFPSPs for different UTs to mitigate the interference between any two different UTs.

To measure the overlap degree of the TB domain channels of any two different UTs, the following parameter is introduced

$$\eta\left(\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u},\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}\right)\triangleq\frac{\mathrm{tr}\left\{\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u}\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}\right\}}{\left\|\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u}\right\|_{\mathrm{F}}\left\|\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u}\right\|_{\mathrm{F}}},\quad(26)$$

From Cauchy-Schwarz inequality, $0\leq\eta\left(\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u},\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}\right)\leq 1$. Since for $\forall u,u'\in\mathcal{U}$, $u\neq u'$, the diagonal elements in $\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u}$ and $\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}$ are all non-negative, $\eta\left(\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u},\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}\right)=0$ if and only if (25) holds.

With the parameter defined in (26), the TFPSP scheduling problem can be formulated as

$$\min_{\{\phi_u,\varphi_u:u\in\mathcal{U}\}}\sum_{u\in\mathcal{U}}\sum_{u'\in\mathcal{U}\setminus\{u\}}\eta\left(\mathbf{R}^{\mathrm{TB}}_{u,\phi_u,\varphi_u},\mathbf{R}^{\mathrm{TB}}_{u',\phi_{u'},\varphi_{u'}}\right).\quad(27)$$

This is a combinational optimization problem, for which we propose a simple approach to find a near-optimal solution. We begin by introducing a threshold value $\gamma$. If the overlap degree between the TB domain channels of two UTs is less than $\gamma$, we can approximate the two UTs as non-overlapping in the TB domain, allowing them to reuse the same TFPSP.

Motivated by [9], we use a graph to divide the UTs into different groups according to their overlap degrees. All the UTs in the system and their mutual overlap degrees can be decribed in an undirected weighted graph $G(\mathcal{U},\mathcal{D})$, where all the UTs are expressed as the vertices and the edges $\mathcal{D}=\left\{\eta\left(\mathbf{R}^{\mathrm{TB}}_{u},\mathbf{R}^{\mathrm{TB}}_{u'}\right)|u,u'\in\mathcal{U},\eta\left(\mathbf{R}^{\mathrm{TB}}_{u},\mathbf{R}^{\mathrm{TB}}_{u'}\right)>\gamma\right\}$ of the graph. The detailed procedures of the whole pilot scheduling algorithm are shown in Algorithm 1, where a smaller $\gamma$ corresponds to a better performance as well as a larger computation overhead.

**Algorithm 1** Pilot Scheduling Algoritm with TFPSPs

**Require:** The UT set $\mathcal{U}$; $\mathbf{R}^{\mathrm{TB}}_{u}$ for $\forall u\in\mathcal{U}$; $G(\mathcal{U},\mathcal{D})$; $\gamma$;
**Ensure:** The phase shift factors $\varphi_u$ and $\phi_u$ for $\forall u\in\mathcal{U}$;
1: **Step 1: DSatur UT grouping**
2: Initialization: $\forall u\in\mathcal{U}$, $c_u=0$;
3: Start with the vertex $u_0$ with the most connected vertices and color it, i.e., $c_{u_0}=1$;
4: **while** $\exists c_u=0$ **do**
5: Select the uncolored vertex $u$ with the most colors in its neighborhood as the current vertex;
6: Use $\mathcal{T}_u$ as the set of color labels in the $u$-th vertex's neighborhood, and color $u$ as $c_u=\min_{x\in\mathbb{N}_+,x\notin\mathcal{T}_u}x$
7: **end while**
8: $C=\max_{u\in\mathcal{U}}c_u$ and divide $\mathcal{U}$ into $\mathcal{U}_0,\mathcal{U}_1,\cdots,\mathcal{U}_{C-1}$ according to the colored $G(\mathcal{U},\mathcal{D})$.
9: **for** $i=0,1,\cdots,C-1$ **do**
10: $\mathbf{R}^{\mathrm{TB}}_{\mathcal{U}_i}=\sum_{u\in U_i}\mathbf{R}^{\mathrm{TB}}_{u}$;
11: **end for**
12: **Step 2: TFPSP scheduling**
13: Initialization: the set of scheduled UT groups $\mathcal{U}^{\mathrm{sch}}=\{0\}$, the set of unscheduled UT $\mathcal{U}^{\mathrm{uns}}=\{0,1,\cdots,C-1\}\backslash\mathcal{U}^{\mathrm{sch}}$, and $\phi_{\mathcal{U}_0}=\varphi_{\mathcal{U}_0}=0$;
14: **for** each $i\in\mathcal{U}^{\mathrm{uns}}$ **do**
15: Search for $\varphi$, $\phi$ satisfying $\eta\left(\mathbf{R}^{\mathrm{TB}}_{\mathcal{U}_i,\phi_{\mathcal{U}_i},\varphi_{\mathcal{U}_i}},\sum_{i'\in\mathcal{U}^{\mathrm{sch}}}\mathbf{R}^{\mathrm{TB}}_{\mathcal{U}_{i'},\phi_{\mathcal{U}_{i'}},\varphi_{\mathcal{U}_{i'}}}\right)\leq\gamma$ and $\varphi_{\mathcal{U}_i}=\varphi$, $\phi_{\mathcal{U}_{i'}}=\phi$;
16: If the phase shift factors cannot be found in the former step, adopt
$$\{\varphi_{\mathcal{U}_i},\phi_{\mathcal{U}_i}\}=\arg\min_{\{\varphi,\phi\}}\eta\left(\mathbf{R}^{\mathrm{TB}}_{\mathcal{U}_i,\phi_{\mathcal{U}_i},\varphi_{\mathcal{U}_i}},\sum_{i'\in\mathcal{U}^{\mathrm{sch}}}\mathbf{R}^{\mathrm{TB}}_{\mathcal{U}_{i'},\phi_{\mathcal{U}_{i'}},\varphi_{\mathcal{U}_{i'}}}\right);$$
17: For each $u\in\mathcal{U}_i$, $\varphi_u=\varphi_{\mathcal{U}_i}$, $\phi_u=\phi_{\mathcal{U}_i}$;
18: Update $\mathcal{U}^{\mathrm{sch}}\leftarrow\mathcal{U}^{\mathrm{sch}}\cup\{i\}$, $\mathcal{U}^{\mathrm{uns}}\leftarrow\mathcal{U}\backslash\mathcal{U}^{\mathrm{sch}}$.
19: **end for**

## V. Simulation Results

In this section, we present simulation results to validate the effectiveness of our proposed TFPSPs. We utilize the well-established QuaDRiGa model [13] to generate the SFT domain channel vector for each UT, and the scenario is set to 3GPP_38.901_UMa_NLOS [13]. The essential simulation parameters are summarized in Table I. We evaluate the performance in two distinct scenarios involving 48 and 300 UTs, respectively. Following the channel generation, the statistical CSI $\mathbf{R}^{\mathrm{TB}}_{u}$ for each $u\in\mathcal{U}$ is acquired using the method proposed in [14].

TABLE I
Parameter Settings

| Parameter | Value |
|---|---|
| Number of BS antennas, $M$ | 128 |
| Number of UTs, $U$ | 48&300 |
| Carrier frequency $f_{\mathrm{c}}$ | 5.8 GHz |
| Number of subcarriers, $N_{\mathrm{c}}$ | 2048 |
| CP length $N_{\mathrm{g}}$ | 144 |
| Number of valid subcarriers $K$ | 360 |
| Subcarrier spacing $\Delta f$ | 15 kHz |
| Number of slot in each frame, $N_{\mathrm{p}}$ | 8 |
| Number of symbols in each slot, $N_{\mathrm{b}}$ | 14 |
| Velocity of UTs, $v_{\mathrm{speed}}$ | 3 km/h |

We firstly present the normalized MSE (NMSE) performance of our proposed channel estimation with TFPSPs under different fine factor settings in Fig. 3. Due to the high computational complexity of MMSE estimation, we use the information geometry approach in [6] to approach the performance of MMSE estimation with a much lower complexity. The simulation is carried out in the scenario involving 48 UTs, and we consider the case when $T=N_{\mathrm{p}}-1$. The NMSE is defined by

$$\varepsilon_{\mathrm{NMSE}}=\frac{1}{N_{\mathrm{sam}}U}\sum_{n=1}^{N_{\mathrm{sam}}}\sum_{u=0}^{U-1}\frac{\left\|\hat{\mathbf{h}}^{\mathrm{SFT,p}}_{u,T}(n)-\mathbf{h}^{\mathrm{SFT,p}}_{u,T}(n)\right\|_2^2}{\left\|\mathbf{h}^{\mathrm{SFT,p}}_{u,T}(n)\right\|_2^2},\quad(28)$$

where $N_{\mathrm{sam}}$ is the number of channel samples in each OFDM symbol, $\mathbf{h}^{\mathrm{SFT,p}}_{u,T}$ is the $n$-th channel vector sample of the $u$-th UT and $\hat{\mathbf{h}}^{\mathrm{SFT,p}}_{u,T}$ is its estimate. It is evident that setting $F_\vartheta=F_\tau=F_\nu=2$ results in a significant performance improvement compared to the case where $F_\vartheta=F_\tau=F_\nu=1$. Moreover, the NMSE with $F_\vartheta=F_\tau=F_\nu=4$ is also lower than that with $F_\vartheta=F_\tau=F_\nu=2$, despite a small gap between them.



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.

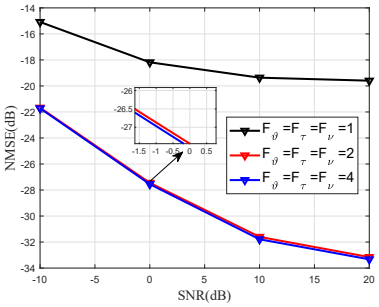



Fig. 3. NMSE of the proposed channel acquition approach under diffenrent fine factor settings, $U = 48$.

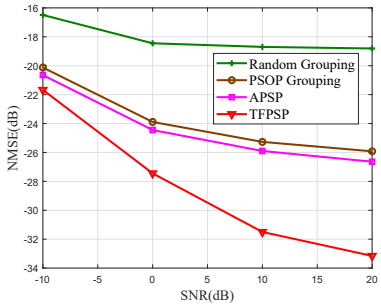



Fig. 4. NMSE under different pilot designs versus SNR, $U = 48$.

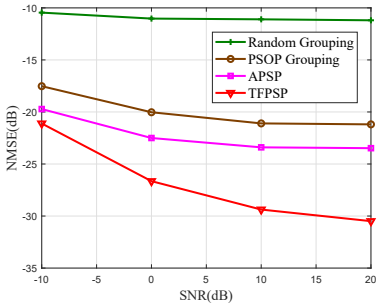



Fig. 5. NMSE under different pilot designs versus SNR, $U = 300$.

We then compare the NMSE performances under different pilot designs. We set $F_\vartheta = F_\tau = F_\nu = 2$ in the folllowing simulations. The following pilot design are compared:

- *Random grouping*: All the UTs are divided into different groups randomly, and UTs in the same group are assigned the same PSOP.
- *Algorithm in* [11]: All the UTs are divided into different groups by the algorithm in [11], and UTs in the same group are assigned the same PSOP.
- *APSP*: APSPs are scheduled by the proposed algorithm in [8].
- *TFPSP*: TFPSPs are scheduled by Algorithm 1.

The NMSE performances under different pilot designs in the scenario involving 48 UTs are shown in Fig. 4. As Fig. 4 shows, compared to other pilot designs, TFPSPs fully exploit the channel sparsity along triple beams and schedule pilots along two dimensions, resulting in lower overlap among the equivalent TB channels and thus achieving better performance.

We further extend the channel acquisition approaches to a scenario involving 300 UTs. As the number of UTs increases, mitigating interference among them becomes more challenging. The results in Fig. 5 show that, compared with other pilot designs, TFPSP still yields much more accurate results. By scheduling pilots in two dimensions simultaneously, TFPSPs make it easier to mitigate overlaps among different UTs, enhancing the accuracy of the channel estimation.

## VI. Conclusion

In this paper, we proposed a TFPSP design for massive MIMO-OFDM channel estimation. We first presented a TB-based channel model. By exploiting the channel characteristics in the TB domain, we developed the TFPSPs, a new pilot design that simultaneously transmits pilot signals in both time and frequency domains. Then we derived the optimal conditions for TFPSPs and proposed the corresponding pilot scheduling algorithm. Simulation results validated the efficiency of our proposed TFPSP approach.

## References

[1] C.-X. Wang, X. H. You, X. Q. Gao *et al.*, "On the road to 6G: Visions, requirements, key technologies, and testbeds," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 905–974, 2nd Quart 2023.

[2] X. H. You, C.-X. Wang, J. Huang, X. Q. Gao, Z. Zhang, M. Wang, Y. Huang, C. Zhang, Y. Jiang, J. Wang *et al.*, "Towards 6G wireless communication networks: Vision, enabling technologies, and new paradigm shifts," *Sci. China Inf. Sci.*, vol. 64, pp. 1–74, Jan. 2021.

[3] C.-X. Wang, J. Huang, H. Wang, X. Q. Gao, X. H. You, and Y. Hao, "6G wireless channel measurements and models: Trends and challenges," *IEEE Veh. Technol. Mag.*, vol. 15, no. 4, pp. 22–32, Dec. 2020.

[4] X. Liu, W. Wang, X. Song, X. Q. Gao, and G. Fettweis, "Sparse channel estimation via hierarchical hybrid message passing for massive MIMO-OFDM systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 11, pp. 7118–7134, Nov 2021.

[5] K.-X. Li, X. Q. Gao, and X.-G. Xia, "Channel estimation for LEO satellite massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7537–7550, Nov. 2023.

[6] J. Yang, Y. Chen, A.-A. Lu, W. Zhong, X. Q. Gao, X. H. You, X.-G. Xia, and D. Slock, "Simplified information geometry approach for massive MIMO-OFDM channel estimation–Part I: Algorithm and fixed point analysis," *arXiv preprint arXiv:2401.02035*, 2024.

[7] L. You, X. Q. Gao, X.-G. Xia, N. Ma, and Y. Peng, "Pilot reuse for massive MIMO transmission over spatially correlated rayleigh fading channels," *IEEE Trans. Wireless Commun.*, vol. 14, no. 6, pp. 3352–3366, Jun. 2015.

[8] L. You, X. Q. Gao, A. L. Swindlehurst, and W. Zhong, "Channel acquisition for massive MIMO-OFDM with adjustable phase shift pilots," *IEEE Trans. Signal Process.*, vol. 64, no. 6, pp. 1461–1476, Mar. 2016.

[9] J. Tang, X. Q. Gao, L. You, D. Shi, J. Yang, X.-G. Xia, X. Zhao, and P. Jiang, "Massive MIMO-OFDM channel acquisition with time-frequency phase-shifted pilots," *preprint*, Mar. 2024.

[10] W. Shen, L. Dai, J. An, P. Fan, and R. W. Heath, "Channel estimation for orthogonal time frequency space (OTFS) massive MIMO," *IEEE Trans. Signal Process.*, vol. 67, no. 16, pp. 4204–4217, Aug. 2019.

[11] D. Shi, L. Song, W. Zhou, X. Q. Gao, C.-X. Wang, and G. Ye Li, "Channel acquisition for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4074–4089, Jun. 2023.

[12] D. Chu, "Polyphase codes with good periodic correlation properties (corresp.)," *IEEE Trans. Inf. Theory*, vol. 18, no. 4, pp. 531–532, Jul. 1972.

[13] S. Jaeckel, L. Raschkowski, K. Börner, and L. Thiele, "QuaDRiGa: A 3-D multi-cell channel model with time evolution for enabling virtual field trials," *IEEE Trans. Antennas Propag.*, vol. 62, no. 6, pp. 3242–3256, Jun. 2014.

[14] M. Al-Shoukairi, P. Schniter, and B. D. Rao, "A GAMP-based low complexity sparse Bayesian learning algorithm," *IEEE Trans. Signal Process.*, vol. 66, no. 2, pp. 294–308, Jan. 2018.



Authorized licensed use limited to: Southeast University. Downloaded on August 04,2025 at 11:32:43 UTC from IEEE Xplore. Restrictions apply.